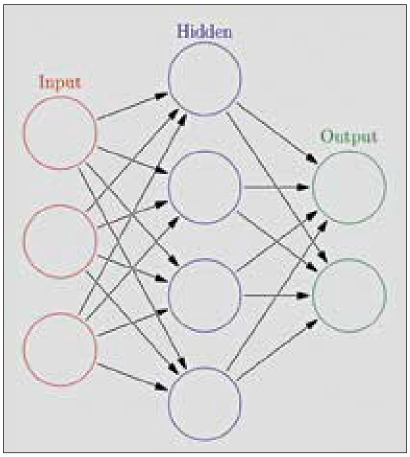
move can be evaluated by comparing the 'value' of eventual outcomes, and thus the machine can indeed learn to play and even master chess. Of course, this task is not as easy as it is being made out to be in this description, but there's an interesting takeaway from all of this.

Machines today can teach themselves, given what they have to learn is in a 'language' they understand. The language being data in terms of numbers, further broken into bytes and bits of ones and zeros. It is here where we can pose the question: Can human emotion be represented by a matrix of numbers? Can we truly encapsulate the intricacy and subjectivity associated with something as elemental as emotion in a number? Also, do we really need to?

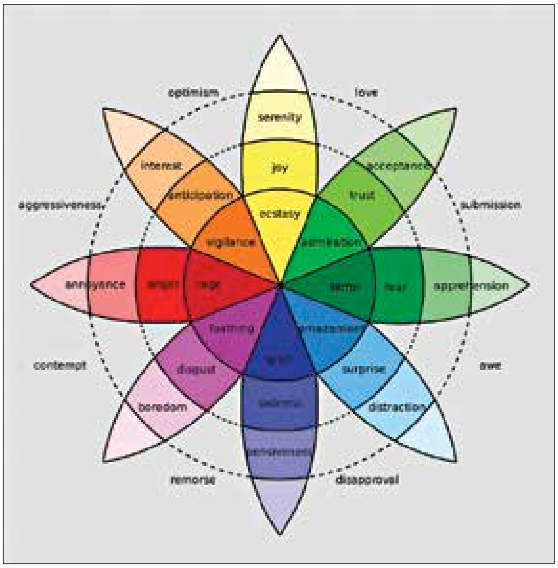


A schematic of the node representing neurons in a simple neural network architecture

What's a machine's EQ?

Emotional Intelligence, usually shortened as EQ is broadly classified as the ability to recognise your own and other people's emotions, to guide our thinking and level of interaction. For machines to have emotional intelligence they must be able to read and differentiate between emotions and respond correctly. A technique called natural language processing allows machines to interpret text and understand its meaning. Image processing tools have gone hand in hand in development of massive computing and gives machines the ability to 'see',

But what about the subtext? What about an underlying emotion that leads to a certain input? Unfortunately, machines are no good at that. Atleast not yet. For a machine to be classified as an emotionally intelligent one it must be able to not only interpret or read an input but also get an estimate of what the input signifies and what makes it different from other inputs. Humans are adept at sensing emotion. When we get a reply - "No!" from a person, our brain registers it as a word from the English language that means something, but we also understand the difference between a no said in exasperation and one that is a curt refusal. Our response is according to to the emotion we recognise. Unlike language which has been shown to follow certain structures, or images which are essentially matrices, emotions are not simple constructs and are thus not easily tenable to the mathematical treatment that scientists so much like to mete out.



Psychologist Robert Plutchik's wheel of emotion, an effort to simplify the concept of emotion

So, therefore, is the quest to develop emotionally intelligent machines a hopeless one? Maybe not.

Can today's machines grasp emotion?

The advances made in computer vision, clubbed with decades of research in artificial intelligence has made the dream to achieve emotional intelligent machines, dubbed affective computing, a possible one. Image recognition softwares today are powerful and smart enough to recognize traditional markers of emotion like smiles, sulks and frowns. These facial features, often general in nature are then tracked and mapped onto a certain set of emotions. This sort of non verbal expression tracking has indeed been successfully used to study how receptive students are to MOOCs and garner feedback to improve online course content.

Psychiatric studies have successfully managed to track depression in patients using the very same formula of non-verbal behaviour tracking. The relevance of using facial recognition goes beyond the specific applications mentioned here. If systems can recognise emotions by facial tracking it would lead to novel methods for medical diagnostics. Educational technology, especially for young kids who are more emotive, can be bolstered with animations being designed for real time changes keeping in sync with the child's emotions.

Emotions can also be tracked from voice data. Human speech has a plethora of markers for emotion: pitch, volume and tone being some of them. While a very basic level of voice based emotion tracking has existed in the form of tele-support systems, recent advances in machine learning has made it possible to develop more sophisticated and robust systems. The possibilities that open up with voice based emotional tracking are numerous to say the least. Voice based assistants in our phones can be transformed from a useful to indispensable tool. Imagine a scenario where Siri is not only able to tell you the nearest hospital but also sense the urgency in your tone and take necessary action. With the kind of rapid development seen in mobile computing hardware it is not too farfetched to envision a future in which a humble phone comes with a